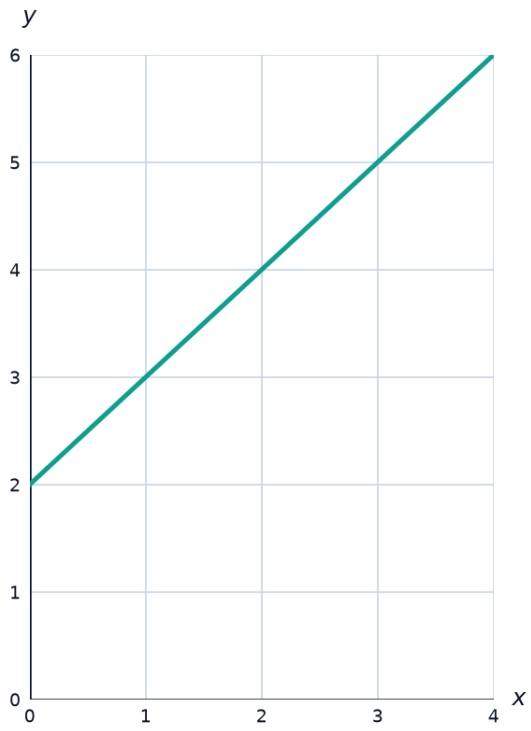
Pre-Calculus Concepts

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Limits: an introductory numerical approach**

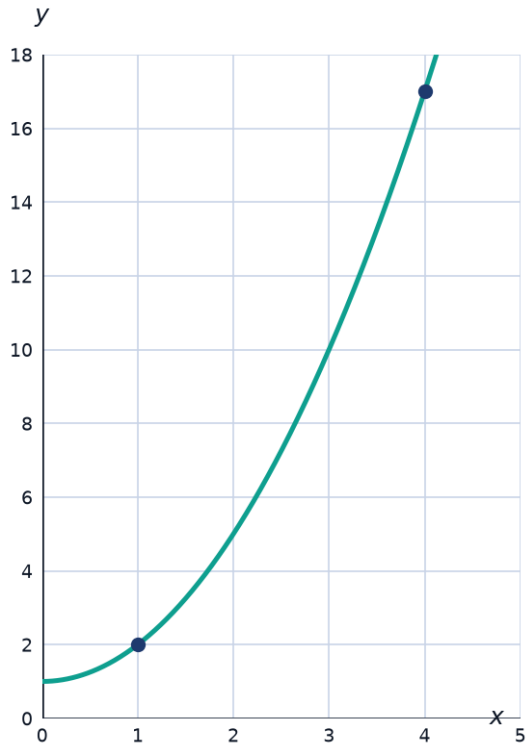
Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.



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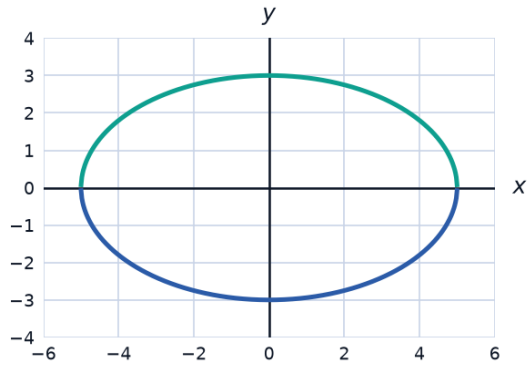
2 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.



3 [Foundational] • Target: Conic sections: ellipses

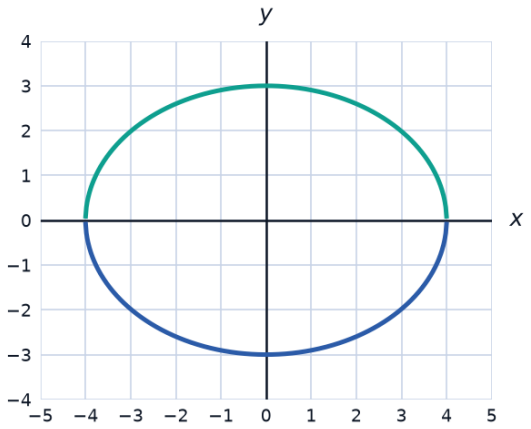
Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).



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4 [Foundational] • Target: Conic sections: ellipses

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.

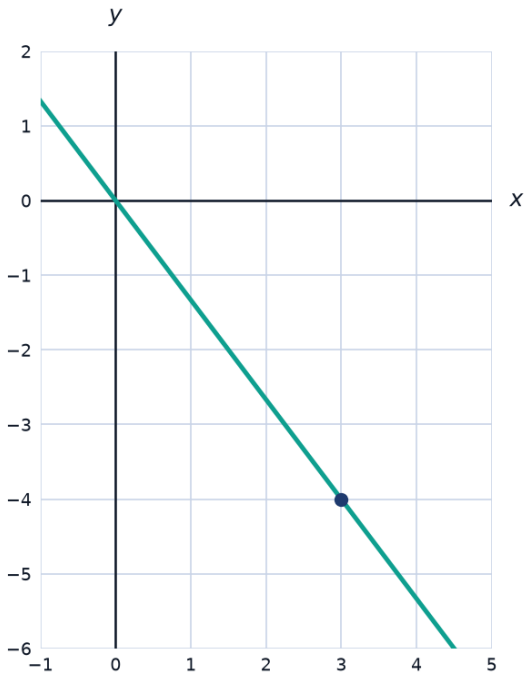


5 [Foundational] • Target: Conic sections: hyperbolas

Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.

6 [Foundational] • Target: Vectors in the plane

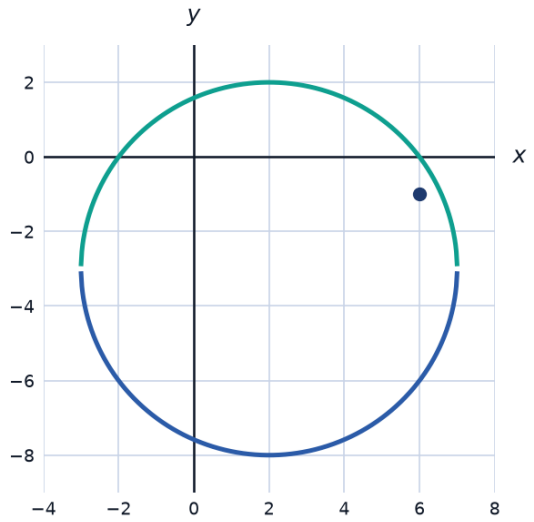
Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.



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7 [Foundational] • Target: A multi-part conic and function analysis problem

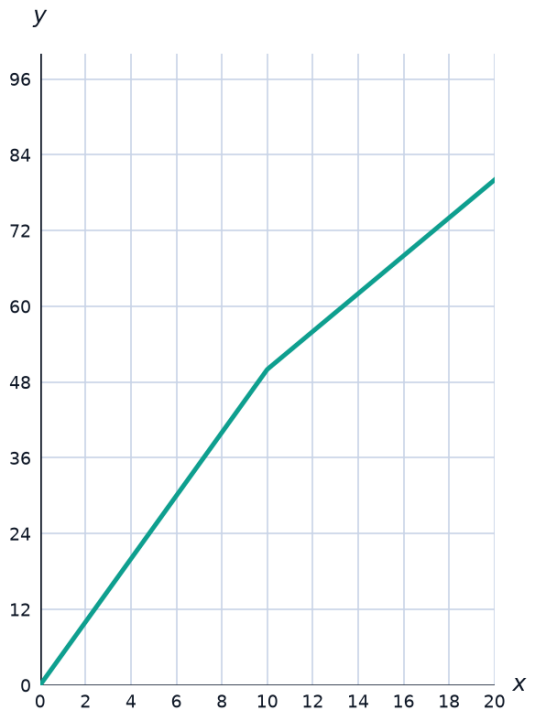
This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise and step functions in applied contexts

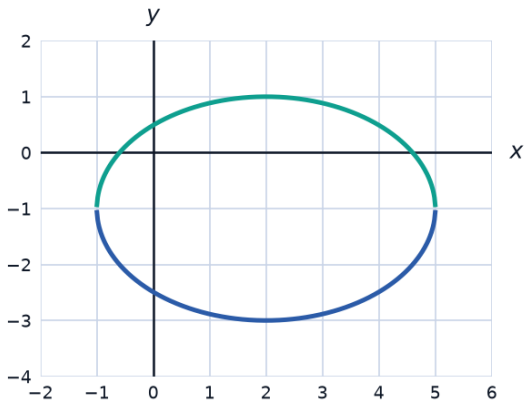
A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.



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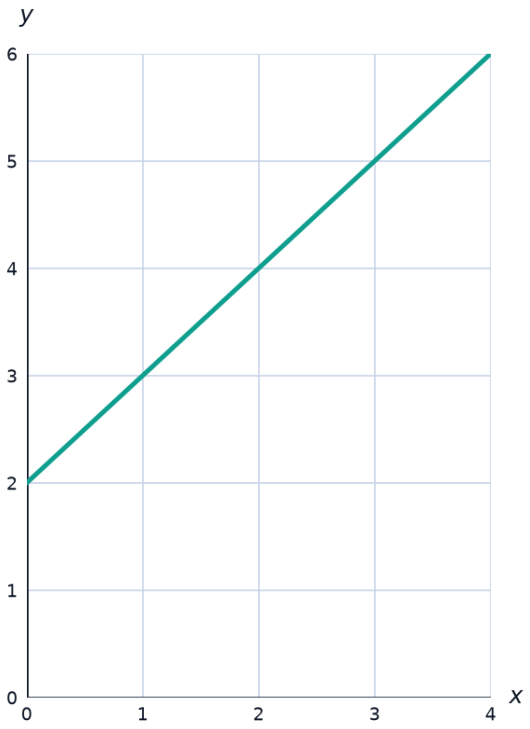
9 [Intermediate] • Target: A multi-part conic section identification and analysis problem

This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.



10 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

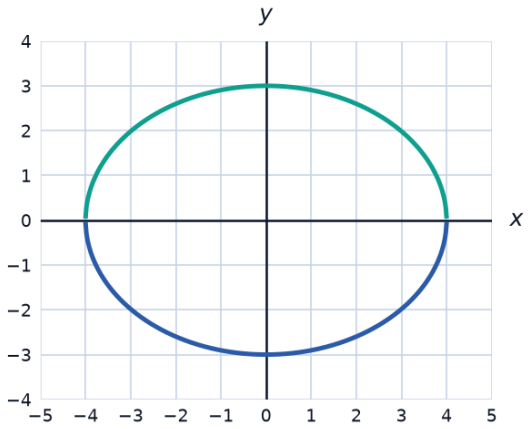
Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.



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11 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.



12 [Intermediate] • Target: Parametric equations

Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x.

13 [Intermediate] • Target: Sequences and series: sigma notation

Evaluate $\sum_{k=1}^{5} (2k + 1)$.

14 [Intermediate] • Target: Sequences and series: sigma notation

Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Binomial theorem • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Use the binomial theorem to expand $(x + 2)^4$.

16 [Challenging] • Target: Binomial theorem • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the coefficient of x^3 in the expansion of $(x + 3)^5$, using $\binom{5}{k}$.

17 [Challenging] • Target: Matrix operations • Cross-Lesson: 8.01 Statistics, Counting & Probability

Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.

18 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Advanced Number Theory

Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \cdots$, using $S = \frac{a_1}{1 - r}$.

19 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 2.01 Advanced Algebra & Systems

Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \cdots$.

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20 [Challenging] • Target: Finding terms of a sequence defined recursively • Cross-Lesson: 3.01 Advanced Functions

A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$. Find a_4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits: an introductory numerical approach

Find the average rate of change of $f(x) = 2x^2 - 3$ over the interval $[-3, 2]$.

a. -10

b. 0

c. -2

d. -7

22 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = 2x^2 - 7x + 9$ over the interval $[1, 3]$.

a. 3

b. -1

c. 2

d. 1

23 [Foundational] • Target: Conic sections: ellipses

Find the average rate of change of $f(x) = 3x^2 + 8x - 9$ over the interval $[5, 6]$.

a. 41

b. 43

c. 40

d. 44

24 [Foundational] • Target: Conic sections: hyperbolas

Find the average rate of change of $f(x) = x^2 + 7x$ over the interval $[-6, 3]$.

a. 4

b. 5

c. 36

d. -5

25 [Foundational] • Target: Parametric equations

Find the average rate of change of $f(x) = x^2 - 9x + 7$ over the interval $[-3, 4]$.

a. -8

b. -56

c. -15

d. -7

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26 [Foundational] • Target: Sequences and series: sigma notation

Find the average rate of change of $f(x) = -3x^2 + 2x + 9$ over the interval $[0, 1]$.

a. -4

b. -1

c. -2

d. 1

27 [Foundational] • Target: Binomial theorem

Find the average rate of change of $f(x) = x^2 - 7$ over the interval $[-2, 1]$.

a. 0

b. -3

c. -4

d. -1

28 [Foundational] • Target: Vectors in the plane

Evaluate $\sum_{k=1}^{10} (k - 1)$.

a. 46

b. 44

c. 45

d. 55

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A multi-part conic and function analysis problem

Evaluate $\sum_{k=1}^{10} (2k - 2)$.

a. 90

b. 88

c. 100

d. 110

30 [Intermediate] • Target: Matrix operations

Evaluate $\sum_{k=1}^{10} (k + 3)$.

a. 85

b. 84

c. 95

d. 55

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31 [Intermediate] - Target: Infinite geometric series

Evaluate $\sum_{k=1}^{8} (k-5)$.

a. -5

b. 4

c. -4

d. 36

32 [Intermediate] - Target: Piecewise and step functions in applied contexts

Evaluate $\sum_{k=1}^{11} (2k+1)$.

a. 132

b. 154

c. 143

d. 141

33 [Intermediate] - Target: Finding terms of a sequence defined recursively

Evaluate $\sum_{k=1}^{11} (-3k+2)$.

a. -176

b. -173

c. -165

d. -198

34 [Intermediate] - Target: A multi-part conic section identification and analysis problem

Evaluate $\sum_{k=1}^{12} (3k+5)$.

a. 306

b. 291

c. 294

d. 234

35 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 9, 0)$

b. $(0, \pm 40)$

c. $(\pm 40, 0)$

d. $(\pm 41, 0)$

36 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 4, 0)$

b. $(\pm 3, 0)$

c. $(0, \pm 4)$

d. $(\pm 5, 0)$

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37 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the ellipse $\frac{x^2}{1369} + \frac{y^2}{144} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 12, 0)$

b. $(\pm 37, 0)$

c. $(0, \pm 35)$

d. $(\pm 35, 0)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

A hyperbola has vertices at $(\pm 3, 0)$ and $b = 4$. Find the coordinates of the foci.

a. $(\pm 5, 0)$

b. $(\pm 7, 0)$

c. $(\pm 3, 0)$

d. $(\pm 4, 0)$

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A hyperbola has vertices at $(\pm 12, 0)$ and $b = 35$. Find the coordinates of the foci.

a. $(\pm 12, 0)$

b. $(\pm 35, 0)$

c. $(\pm 37, 0)$

d. $(\pm 47, 0)$

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1 - r}$.

a. $\frac{55}{4}$

b. $\frac{44}{5}$

c. -44

d. $\frac{44}{3}$

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41 [Challenging] • Target: Limits: an introductory numerical approach • Cross-Lesson: 4.01 Advanced Trigonometry

Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{4}$, using $S = \frac{a_1}{1 - r}$.

a. 9

b. 48

c. $\frac{48}{5}$

d. 16

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

For the ellipse $\frac{x^2}{841} + \frac{y^2}{400} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 29, 0)$

b. $(0, \pm 21)$

c. $(\pm 21, 0)$

d. $(\pm 20, 0)$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the sum of the infinite geometric series with first term $a_1 = 11$ and common ratio $r = -\frac{1}{4}$, using $S = \frac{a_1}{1 - r}$.

a. $\frac{55}{4}$

b. $\frac{44}{5}$

c. -44

d. $\frac{44}{3}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Statistics, Counting & Probability

In an arithmetic sequence, the 4th term is 19 and the 11th term is 54. What is the first term a_1?

a. 4

b. 5

c. 6

d. 7

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

What is the 4th term in the expansion of $(x + 2)^5$ arranged in descending powers of x?

a. $40x^2$

b. $80x^2$

c. $20x^3$

d. $10x^3$

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